CAMBRIDGE
International Education

Cambridge O Level

PHYSICS **5054/21**

Paper 2 Theory **October/November 2024**

MARK SCHEME

Maximum Mark: 75

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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Question	Answer	Marks
1(a)	(m =) ρV **or** 650 × 0.0012 **or** 0.78	**C1**
	(combined mass =) 0.98 (kg)	**A1**
1(b)(i)	(F =) kx **or** 14 × 0.035	**C1**
	0.49 (N)	**A1**
1(b)(ii)	(W =) Fx **or** 0.49 × 0.86	**C1**
	0.42 (J)	**A1**
1(b)(iii)	the distance moved / work done (in one second / unit time / a given time) increases (as the speed increases)	**B1**
	the work done in one second / unit time increases	**B1**

Question	Answer	Marks
2(a)(i)	(G.P.E.=) mgh **or** 0.20 × 9.8 × 40	**C1**
	78 (J)	**A1**
2(a)(ii)	(K.E. =) ½mv^2 or ½ × 0.2 × v^2	**C1**
	(v =) $\sqrt{2gh}$ **or** $\sqrt{2 \times 9.8 \times 40}$ **or** $\sqrt{784}$	**C1**
	28 (m / s)	**A1**
2(b)(i)	($\Delta\theta$ =) ΔE / mc **or** 78.4 / (0.20 × 1400) **or** gh / c **or** 9.8 × 40 / 1400	**C1**
	0.28 (°C)	**A1**
2(b)(ii)	some (thermal) energy is transferred to the ground **or** work is done deforming the modelling clay **or** air resistance / work done means that speed / k.e. is less than the calculated value	**B1**

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Question	Answer	Marks
3(a)	vibrating atoms / particles (**ignore** electrons) collide with electrons	**B1**
	electrons move (a long distance) through metal	**B1**
	(moving) electrons collide / interact with distant atoms (**ignore** electrons) **or** electrons carry energy (through metal)	**B1**
3(b)(i)	their / the particles' (average) speed increases	**B1**
3(b)(ii)	the particles collide (with each other / the walls of the container)	**B1**
	they / the particles exert greater impulse / force	**B1**
	collide more often / harder with the walls (of the cylinder)	**B1**
3(c)(i)	(particles are further apart) so fewer collisions in unit time / less frequent collisions / longer time between collisions	**B1**
3(c)(ii)	pressure becomes equal to atmospheric pressure	**B1**
	no resultant force on piston **or** force to the left equals force to the right **or** air resistance / friction (between piston and cylinder walls) mentioned	**B1**

Question	Answer	Marks
4(a)	at least **one** ray from left-hand side showing correct refraction at both faces **and** no incorrect rays	**B1**
	at least **two** correct rays (i.e. two rays within glass) **and** one ending at R and the other at V	**B1**
4(b)	both (red and violet light) travel more slowly in glass than in air	**B1**
	red light travels more quickly (than violet light) in glass	**B1**
4(c)(i)	any **four** (of the five) colours in order **or** five colours in any order	**C1**
	red orange yellow green blue indigo **violet**	**B1**

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Question	Answer	Marks
4(c)(ii)	frequency increases **and** wavelength decreases	**B1**
4(d)	infrared (radiation)	**B1**

Question	Answer	Marks
5(a)	ultrasound / sound / it is transmitted by <u>vibrating</u> matter / particles	**B1**
	no matter / particles in a vacuum / no medium	**B1**
5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	**B1**
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	**B1**
5(c)(i)	(λ =) v / f **or** 1500 / 8.4×10^6	**C1**
	1.8×10^N	**C1**
	1.8×10^{-4} (m)	**A1**
5(c)(ii)	frequency: does not change	**B1**
	wavelength: increases	**B1**

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Question	Answer	Marks
6(a)	steel	**B1**
6(b)	at least **two complete** field lines on each side of magnet	**B1**
	at least **one** field line on each side that begins and ends on short side / corner of magnet and all lines drawn of correct shape	**B1**
	at least one arrowhead from N to S **and** no arrowheads wrong	**B1**
6(c)(i)	one arrow at each pole in opposite directions **or** one arrow at each pole parallel to field lines	**C1**
	one horizontal arrow pointing left at S pole **and** one horizontal arrow pointing right at N pole **and** arrows parallel to field lines	**A1**
6(c)(ii)	forces produce moment / turning effect /couple / torque	**B1**
	magnet / it rotates	**B1**
	until it is / finishes parallel to the field lines / horizontal **or** rotates anticlockwise	**B1**

Question	Answer	Marks
7(a)(i)	(I =) P / V or 24 / 12	**C1**
	2.0 (A)	**A1**
7(a)(ii)	(R =) V / I **or** 12 / 2.0	**C1**
	6.0 (Ω)	**A1**

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Question	Answer	Marks
7(b)(i)	variable resistor in series with battery and lamp	**B1**
	ammeter in series with battery and lamp **and** voltmeter across lamp	**B1**
	or	
	potential divider circuit correctly connected	**B1**
	ammeter in series with lamp and voltmeter across lamp	**B1**
7(b)(ii)	curved line with gradient ⩾ 0 **and** starting at origin	**M1**
	curve of decreasing gradient and gradient always > 0	**A1**
7(b)(iii)	resistance increases	**B1**
	temperature increases/filament gets hot **or** particles / ions vibrate more vigorously (and impede electron flow)	**B1**

Question	Answer	Marks
8(a)	(a nucleus of hydrogen–3) contains only 3 nucleons **or** only 1 proton	**B1**
8(b)(i)	${}^{0}_{-1}\beta$	**B1**
	${}^{3}Q$	**B1**
	${}_{2}Q$	**B1**
8(b)(ii)	helium	**B1**
8(c)(i)	<u>time</u> taken for something to halve	**C1**
	<u>time</u> taken for number of atoms / (corrected) count rate to halve	**A1**

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Question	Answer	Marks
8(c)(ii)	(due to) spontaneous / random nature of decay	**B1**

Question	Answer	Marks
9(a)(i)	(nuclear) fusion	**B1**
	hydrogen (nuclei) join / fuse together	**B1**
	to produce helium (nuclei)	**B1**
9(a)(ii)	reaction / it transfers energy thermally	**B1**
	(high temperature) produces (high) pressure / outward force	**B1**
	(inward) gravitational force balanced **or** (further) gravitational collapse prevented	**B1**
9(b)(i)	not enough hydrogen (left) **or** runs out of hydrogen	**B1**
9(b)(ii)	any **three** from: (it / red supergiant) explodes turns into a supernova produces heavy elements forms a nebula leaves behind a neutron star / black hole	**B3**

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